

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics
PRACTICAL NO - 12

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Class: TYIT	Batch:2
Date of Assignment: 3-10-26	Date/Time of Submission: 3-10-26

Practical 12: Analysis of Variance and Experimental Design

12. Analysis of Variance and Experimental Design:

a. Design and conduct an experiment to study the effects of different factors on a specific response variable, analyze the results using analysis of variance (ANOVA), and draw conclusions.

CODE:-

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import f_oneway
import statsmodels.api as sm
from statsmodels.formula.api import ols
```

LOAD DATASET

```
file_path = "Retail and warehouse Sale - Retail and warehouse Sale.csv"
df = pd.read_csv(file_path)
print("Dataset loaded successfully!")
```

PART A – ANALYSIS OF VARIANCE (ANOVA)

1. Display first five records

```
print("\n1. First Five Records:")
print(df.head())
```

2. Number of rows and columns

```
print("\n2. Dataset Shape:")
print("Number of Rows:", df.shape[0])
print("Number of Columns:", df.shape[1])
```

3. Column names and data types

```
print("\n3. Column Names and Data Types:")
print(df.dtypes)
```

4. Missing values and duplicate records

```
print("\n4. Missing Values:")
print(df.isnull().sum())
print("\nNumber of Duplicate Records:")
print(df.duplicated().sum())
```

5. Different ITEM TYPE categories

```
print("\n5. Different ITEM TYPE Categories:")
print(df["ITEM TYPE"].unique())
print("\nNumber of ITEM TYPE Categories:")
print(df["ITEM TYPE"].nunique())
```

6. Average RETAIL SALES for each ITEM TYPE

```
average_sales = df.groupby("ITEM TYPE")["RETAIL SALES"].mean()
print("\n6. Average RETAIL SALES for Each ITEM TYPE:")
print(average_sales)
```

7. Box plot

```
plt.figure(figsize=(12, 6))

sns.boxplot(
    x="ITEM TYPE",
    y="RETAIL SALES",
    data=df
)
plt.title("Retail Sales by Item Type")
plt.xlabel("Item Type")
plt.ylabel("Retail Sales")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

8. Hypotheses

```
print("\n8. Hypotheses:")
print("H0: The average RETAIL SALES are equal for all ITEM TYPE categories.")
print("H1: At least one ITEM TYPE has a different average RETAIL SALES.")
```

9 & 10. One-way ANOVA

```
groups = [
    group["RETAIL SALES"].dropna()
    for name, group in df.groupby("ITEM TYPE")
]
f_statistic, p_value = f_oneway(*groups)
print("\n9 & 10. One-Way ANOVA Results:")
print("F-statistic:", f_statistic)
print("p-value:", p_value)
```

11 & 12. Compare p-value with 0.05

```
alpha = 0.05
print("\n11 & 12. Hypothesis Test:")

if p_value < alpha:
    print("p-value < 0.05")
    print("Reject the Null Hypothesis (H0).")
else:
    print("p-value >= 0.05")
    print("Do Not Reject the Null Hypothesis (H0).")
```

13. Conclusion

```
print("\n13. Conclusion:")
if p_value < alpha:
    print(
        "There is a statistically significant difference in average "
        "RETAIL SALES among the different ITEM TYPE categories."
    )
else:
```

```

print(
    "There is no statistically significant difference in average "
    "RETAIL SALES among the different ITEM TYPE categories."
)
print("Khushi T032")

```

OUTPUT:-

Dataset loaded successfully!

1. First Five Records:

	YEAR	MONTH	SUPPLIER	ITEM CODE
0	2020	1	REPUBLIC NATIONAL DISTRIBUTING CO	100009
1	2020	1	PWSWN INC	100024
2	2020	1	RELIABLE CHURCHILL LLLP	1001
3	2020	1	LANTERNA DISTRIBUTORS INC	100145
4	2020	1	DIONYSOS IMPORTS INC	100293

	ITEM DESCRIPTION	ITEM TYPE	RETAIL SALES
0	BOOTLEG RED - 750ML	WINE	0.00
1	MOMENT DE PLAISIR - 750ML	WINE	0.00
2	S SMITH ORGANIC PEAR CIDER - 18.7OZ	BEER	0.00
3	SCHLINK HAUS KABINETT - 750ML	WINE	0.00
4	SANTORINI GAVALA WHITE - 750ML	WINE	0.82

	RETAIL TRANSFERS	WAREHOUSE SALES
0	0.0	2.0
1	1.0	4.0
2	0.0	1.0
3	0.0	1.0
4	0.0	0.0

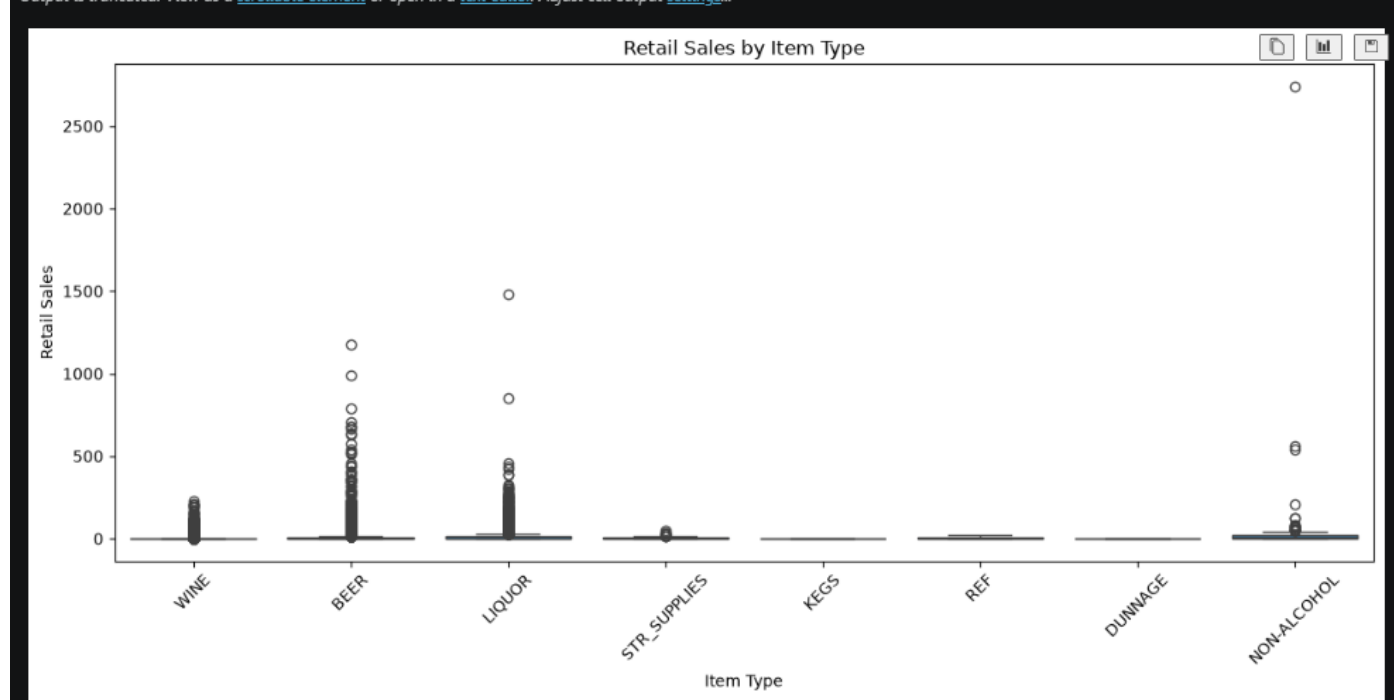
2. Dataset Shape:

```

...
REF          5.783750
STR_SUPPLIES 5.485714
WINE         3.195334
Name: RETAIL SALES, dtype: float64

```

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```

8. Hypotheses:
H0: The average RETAIL SALES are equal for all ITEM TYPE categories.
H1: At least one ITEM TYPE has a different average RETAIL SALES.

9 & 10. One-Way ANOVA Results:
F-statistic: 123.52155338346167
p-value: 8.685629095379197e-180

11 & 12. Hypothesis Test:
p-value < 0.05
Reject the Null Hypothesis (H0).

13. Conclusion:
There is a statistically significant difference in average RETAIL SALES among the different ITEM TYPE categories.
Khushi T032

```

b. Implement a factorial experiment and analyze the main effects and interaction effects of factors using statistical techniques.

CODE:-

```

# PART B – FACTORIAL EXPERIMENT
print("\n\n=====")
print("PART B – FACTORIAL EXPERIMENT")
print("=====")

# 14. Select ITEM TYPE and YEAR as factors
print("\n14. Factors:")
print("Factor 1: ITEM TYPE")
print("Factor 2: YEAR")
print("Response Variable: RETAIL SALES")

# Check available years
print("\nAvailable Years:")
print(sorted(df["YEAR"].dropna().unique()))

print("\nNumber of Different Years:",
      df["YEAR"].nunique())

# 15. Average retail sales for each combination
combination_average = (
    df.groupby(["ITEM TYPE", "YEAR"])["RETAIL SALES"]
    .mean()
    .reset_index()
)
print("\n15. Average RETAIL SALES for Each ITEM TYPE and YEAR:")
print(combination_average)

# Pivot table for better display
pivot_table = pd.pivot_table(
    df,
    values="RETAIL SALES",
    index="ITEM TYPE",
    columns="YEAR",
    aggfunc="mean"
)
print("\nAverage Sales Pivot Table:")

```

```
print(pivot_table)
```

16. Visualization

```
plt.figure(figsize=(12, 6))
```

```
sns.barplot(
    data=df,
    x="ITEM TYPE",
    y="RETAIL SALES",
    hue="YEAR",
    estimator="mean",
    errorbar=None
)
plt.title("Average Retail Sales by Item Type and Year")
plt.xlabel("Item Type")
plt.ylabel("Average Retail Sales")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

17. Main effect of ITEM TYPE

```
print("\n17. Main Effect of ITEM TYPE:")
```

```
item_type_means = (
    df.groupby("ITEM TYPE")["RETAIL SALES"]
    .mean()
    .sort_values(ascending=False)
)
print(item_type_means)
```

18. Main effect of YEAR

```
print("\n18. Main Effect of YEAR:")
year_means = (
    df.groupby("YEAR")["RETAIL SALES"]
    .mean()
)
print(year_means)
```

19–22. Factorial ANOVA

```
print("\n19–22. Factorial ANOVA:")
if df["YEAR"].nunique() < 2:
    print("\nWARNING:")
    print("The dataset contains only one YEAR:",
          df["YEAR"].unique()[0])
    print(
        "\nFactorial ANOVA cannot be performed using YEAR "
        "because at least two different YEAR levels are required."
    )
    print("\nTherefore:")
    print("ITEM TYPE main effect: Can be analyzed.")
    print("YEAR main effect: Cannot be tested statistically.")
    print("ITEM TYPE × YEAR interaction: Cannot be tested.")
```

```
else:
```

Two-way factorial ANOVA

```

model = ols(
    'Q("RETAIL SALES") ~ C(Q("ITEM TYPE")) * C(YEAR)',
    data=df
).fit()
anova_table = sm.stats.anova_lm(model, typ=2)
print("\n20. Factorial ANOVA Results:")
print(anova_table)
print("\n21. P-values:")
for effect in anova_table.index:
    print(effect, "p-value =",
          anova_table.loc[effect, "PR(>F)"])

```

22. Identify significant effects

```

print("\n22. Significance at alpha = 0.05:")
for effect in anova_table.index:
    if effect == "Residual":
        continue
    p = anova_table.loc[effect, "PR(>F)"]
    if p < 0.05:
        print(effect, "-> Significant")
    else:
        print(effect, "-> Not Significant")

```

23. Final Business Conclusion

```

print("\n23. FINAL BUSINESS CONCLUSION:")
if p_value < 0.05:
    print(
        "The one-way ANOVA indicates that ITEM TYPE has a "
        "significant effect on RETAIL SALES. Therefore, "
        "retail sales vary significantly among different "
        "item categories."
    )
else:
    print(
        "The one-way ANOVA indicates that ITEM TYPE does not "
        "have a statistically significant effect on RETAIL SALES."
    )

if df["YEAR"].nunique() < 2:
    print(
        "The uploaded dataset contains only the year 2020. "
        "Hence, the effect of YEAR and the ITEM TYPE × YEAR "
        "interaction cannot be statistically evaluated."
    )
else:
    print(
        "The factorial ANOVA evaluates the effects of ITEM TYPE, "
        "YEAR, and their interaction on RETAIL SALES."
    )
print("Khushi T032")

```

OUTPUT:-

PART B - FACTORIAL EXPERIMENT

14. Factors:

Factor 1: ITEM TYPE

Factor 2: YEAR

Response Variable: RETAIL SALES

Available Years:

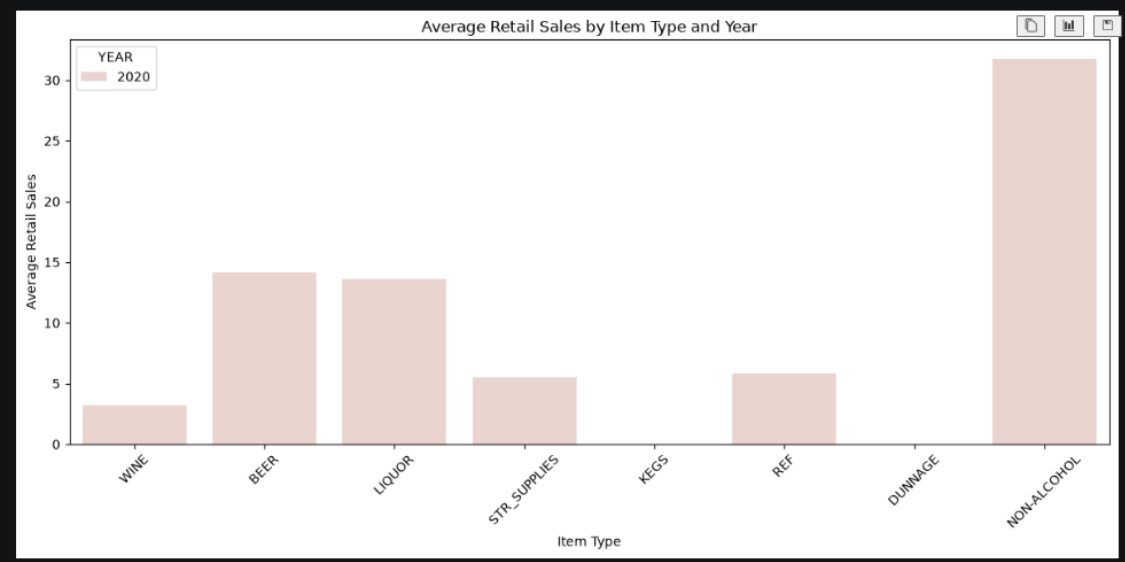
[np.int64(2020)]

Number of Different Years: 1

15. Average RETAIL SALES for Each ITEM TYPE and YEAR:

ITEM TYPE	YEAR	RETAIL SALES
0 BEER	2020	14.118748
1 DUNNAGE	2020	0.000000
2 KEGS	2020	0.000000
3 LIQUOR	2020	13.635171
4 NON-ALCOHOL	2020	31.742419
5 REF	2020	5.783750
6 STR_SUPPLIES	2020	5.485714
...		
NON-ALCOHOL		31.742419
REF		5.783750
STR_SUPPLIES		5.485714
WINE		3.195334

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17. Main Effect of ITEM TYPE:

ITEM TYPE	RETAIL SALES
NON-ALCOHOL	31.742419
BEER	14.118748
LIQUOR	13.635171
REF	5.783750
STR_SUPPLIES	5.485714
WINE	3.195334
KEGS	0.000000
DUNNAGE	0.000000

Name: RETAIL SALES, dtype: float64

18. Main Effect of YEAR:

YEAR	RETAIL SALES
2020	6.939796

Name: RETAIL SALES, dtype: float64

19-22. Factorial ANOVA:

WARNING:

The dataset contains only one YEAR: 2020

Factorial ANOVA cannot be performed using YEAR because at least two different YEAR levels are required.

...

23. FINAL BUSINESS CONCLUSION:

The one-way ANOVA indicates that ITEM TYPE has a significant effect on RETAIL SALES. Therefore, retail sales vary significantly among different item categories. The uploaded dataset contains only the year 2020. Hence, the effect of YEAR and the ITEM TYPE × YEAR interaction cannot be statistically evaluated.

Khushi T032